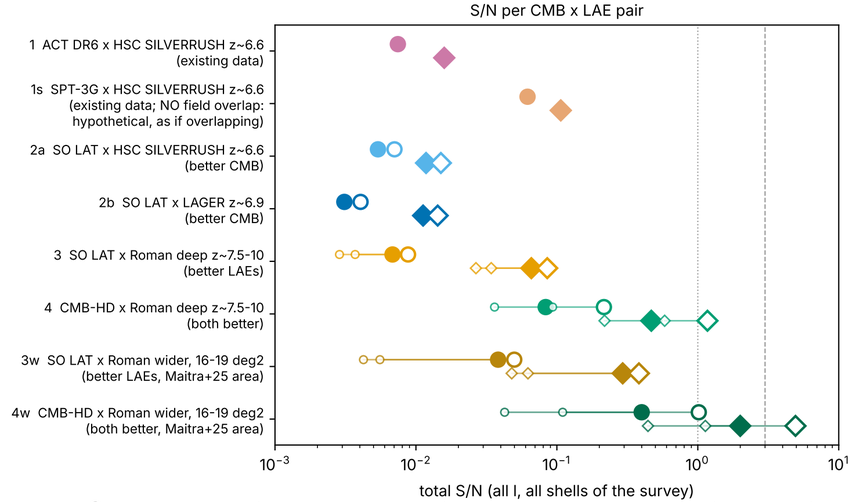
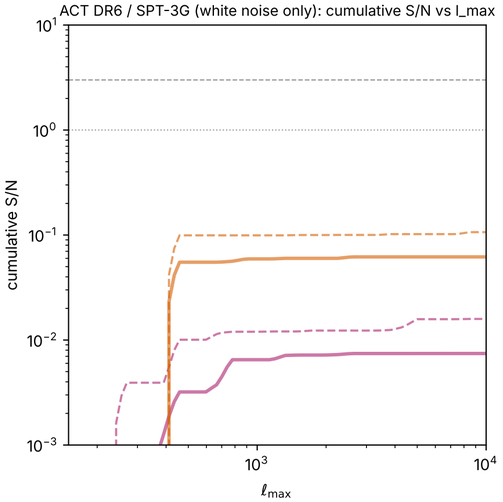
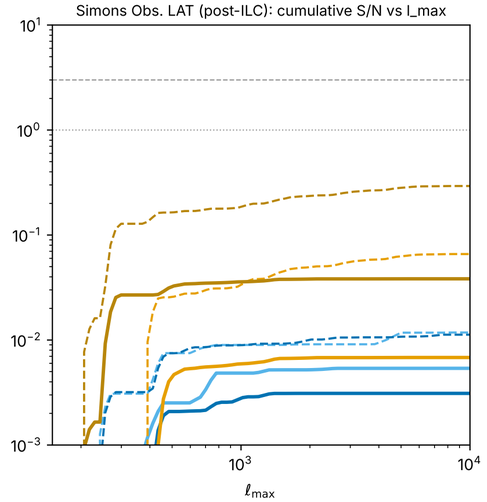
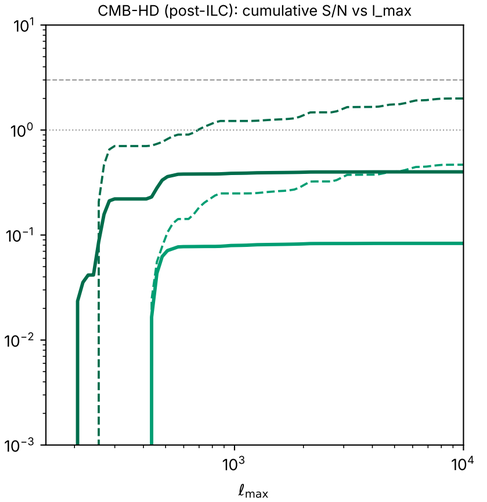
kSZ^2 x LAE detectability for the survey pairs: existing data, better CMB, better LAEs, both (fiducial reionisation history; real areas, counts and footprints)









- La Plante's galaxy model (validated; may be low by up to 2x)
- simulated LAEs (excess not yet confirmed)
- filled: post-ILC noise (ACT / SPT-3G: white noise only)
- open: white (instrument) noise
- small marker: Roman deep: z = 7.3 LF (observed decline); Roman wider: HLWAS deep-tier depth
- 1σ
- 3σ

- 1: ACT DR6 x HSC SILVERRUSH z~6.6, La Plante's galaxy model: 0.0074
- 1: ACT DR6 x HSC SILVERRUSH z~6.6, simulated LAEs: 0.016
- 1s: SPT-3G x HSC SILVERRUSH z~6.6 (no overlap; as if), La Plante's galaxy model: 0.062
- 1s: SPT-3G x HSC SILVERRUSH z~6.6 (no overlap; as if), simulated LAEs: 0.11
- 1σ
- 3σ

- 2a: SO LAT x HSC SILVERRUSH z~6.6, La Plante's galaxy model: 0.0054
- 2a: SO LAT x HSC SILVERRUSH z~6.6, simulated LAEs: 0.012
- 2b: SO LAT x LAGER z~6.9, La Plante's galaxy model: 0.0031
- 2b: SO LAT x LAGER z~6.9, simulated LAEs: 0.011
- 3: SO LAT x Roman deep z~7.5-10, La Plante's galaxy model: 0.0068
- 3: SO LAT x Roman deep z~7.5-10, simulated LAEs: 0.066
- 3w: SO LAT x Roman wider, 16-19 deg2, La Plante's galaxy model: 0.038
- 3w: SO LAT x Roman wider, 16-19 deg2, simulated LAEs: 0.29
- 1σ
- 3σ

- 4: CMB-HD x Roman deep z~7.5-10, La Plante's galaxy model: 0.083
- 4: CMB-HD x Roman deep z~7.5-10, simulated LAEs: 0.47
- 4w: CMB-HD x Roman wider, 16-19 deg2, La Plante's galaxy model: 0.4
- 4w: CMB-HD x Roman wider, 16-19 deg2, simulated LAEs: 2
- 1σ
- 3σ